

Name : Rohit Sharma

Roll No.: 21CS1008

1. The operating system is system software that manages computer hardware and provides common services for programs. It acts as an intermediary between the user and the hardware.
2. A process is an instance of a program in execution and has its own memory space, whereas a thread is the smallest unit of execution within a process and shares the same memory space with other threads of the same process.
- 3.(a) Normalization is the process of organizing data in a database to reduce redundancy and avoid anomalies. It involves dividing larger tables into smaller related tables and defining relationships between them.
- 3.(b) Denormalization can be useful in data warehousing when read performance is more important than storage. By combining tables, complex joins are avoided and queries run faster.
4. A primary key is a unique identifier for a record in a table. It cannot contain NULL values and there can be only one primary key in a table.
5. (No answer)
- 6.(a) Supervised learning uses labelled training data, where each example has an input and the corresponding correct output.
- 6.(b) Unsupervised learning uses data without labels and tries to find patterns or structure in the data.
7. Indexes are used to speed up data retrieval operations on a database table. They reduce the number of disk I/Os by allowing the database to find data without scanning the entire table.

(Continued on next page)

7. (Continued)
However, indexes take extra storage space and slow down write operations like INSERT, UPDATE and DELETE because the index also needs to be updated.
8. Overfitting is when a model learns the training data too well, including the noise, and performs poorly on new unseen data. It can be reduced using techniques like regularization, cross-validation and using more training data.
9. REST API stands for Representational State Transfer Application Programming Interface. It is an architectural style for designing networked applications using HTTP methods.
10. The array is [2, 4, 6, 8, 10].
Mean = $(2+4+6+8+10)/5 = 30/5 = 6$
Sorted array: [2, 4, 6, 8, 10]
Median = 6
- 11.(a) A hash table is a data structure that stores key-value pairs. It uses a hash function to compute an index into an array where the value is stored.
- 11.(b) When two keys produce the same hash value, it is called a collision. It can be resolved using chaining or open addressing.
12. GET is used to retrieve data from the server.
POST is used to send data to the server to create a resource.
PUT is used to update an existing resource.
DELETE is used to remove a resource.